SHETH L.U.J & SIR M.V COLLEGE OF SCIENCE
SUBJECT : AI

Aim: Association Rule Mining
Implement the Association Rule Mining algorithm (e.g., Apriori) to find frequent itemsets.
Generate association rules from the frequent itemsets and calculate their support and confidence.
Interpret and analyze the discovered association rules.

CODE:

```
# ================================================================
# ASSOCIATION RULE MINING USING APRIORI
# ONLINE RETAIL TRANSACTION DATASET
# ================================================================

import pandas as pd
from mlxtend.frequent_patterns import apriori, association_rules

# Load dataset
data = pd.read_csv("Online Retail.csv", encoding="ISO-8859-1")

print("=" * 60)
print("DATASET INFORMATION")
print("=" * 60)
print("Dataset Shape:", data.shape)
print("\nFirst 5 Records:")
print(data.head())

# ------------------------------------------------
# Data Preprocessing
# ------------------------------------------------

# Remove rows with missing invoice numbers or item descriptions
data = data.dropna(subset=["InvoiceNo", "Description"])

# Remove cancelled transactions
data = data[~data["InvoiceNo"].astype(str).str.startswith("C")]

# Keep only positive quantities
data = data[data["Quantity"] > 0]
```

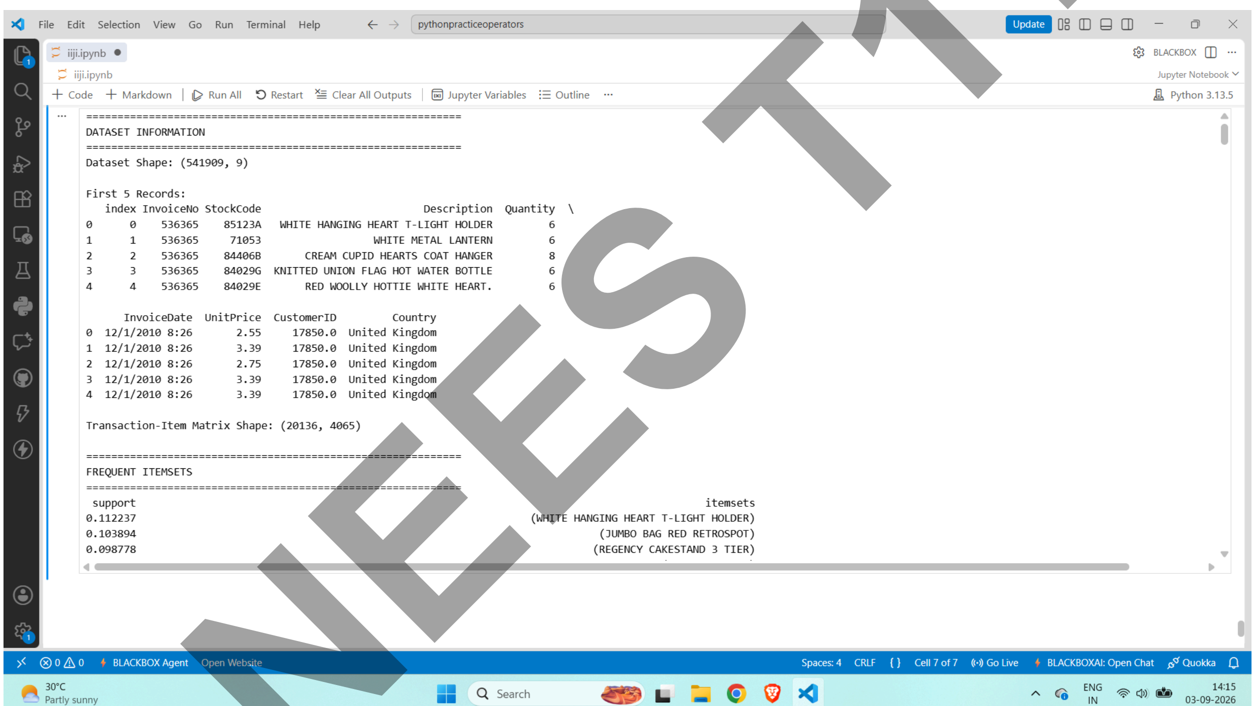
```
# Remove extra spaces from item names
data["Description"] = data["Description"].str.strip()

# --------------------------------------------------
# Create Transaction-Item Matrix
# --------------------------------------------------

basket = (
    data.groupby(["InvoiceNo", "Description"])["Quantity"]
    .sum()
    .unstack()
    .fillna(0)
)

# Convert quantities to True/False
basket = basket > 0

print("\nTransaction-Item Matrix Shape:", basket.shape)

# --------------------------------------------------
# Find Frequent Itemsets using Apriori
# --------------------------------------------------

frequent_itemsets = apriori(
    basket,
    min_support=0.02,
    use_colnames=True
)

frequent_itemsets = frequent_itemsets.sort_values(
    by="support",
    ascending=False
)

print("\n" + "=" * 60)
print("FREQUENT ITEMSETS")
print("=" * 60)

print(frequent_itemsets.to_string(index=False))
```

```
# ------------------------------------------------
# Generate Association Rules
# ------------------------------------------------

rules = association_rules(
    frequent_itemsets,
    num_itemsets=len(basket),
    metric="confidence",
    min_threshold=0.30
)

# Select important columns
rules = rules[
    ["antecedents", "consequents", "support",
     "confidence", "lift"]
]

# Sort rules by confidence
rules = rules.sort_values(
    by="confidence",
    ascending=False
)

print("\n" + "=" * 60)
print("ASSOCIATION RULES")
print("=" * 60)

if len(rules) > 0:
    print(rules.to_string(index=False))
else:
    print("No association rules found.")

# ------------------------------------------------
# Display Top 10 Rules
# ------------------------------------------------

print("\n" + "=" * 60)
print("TOP 10 ASSOCIATION RULES")
```

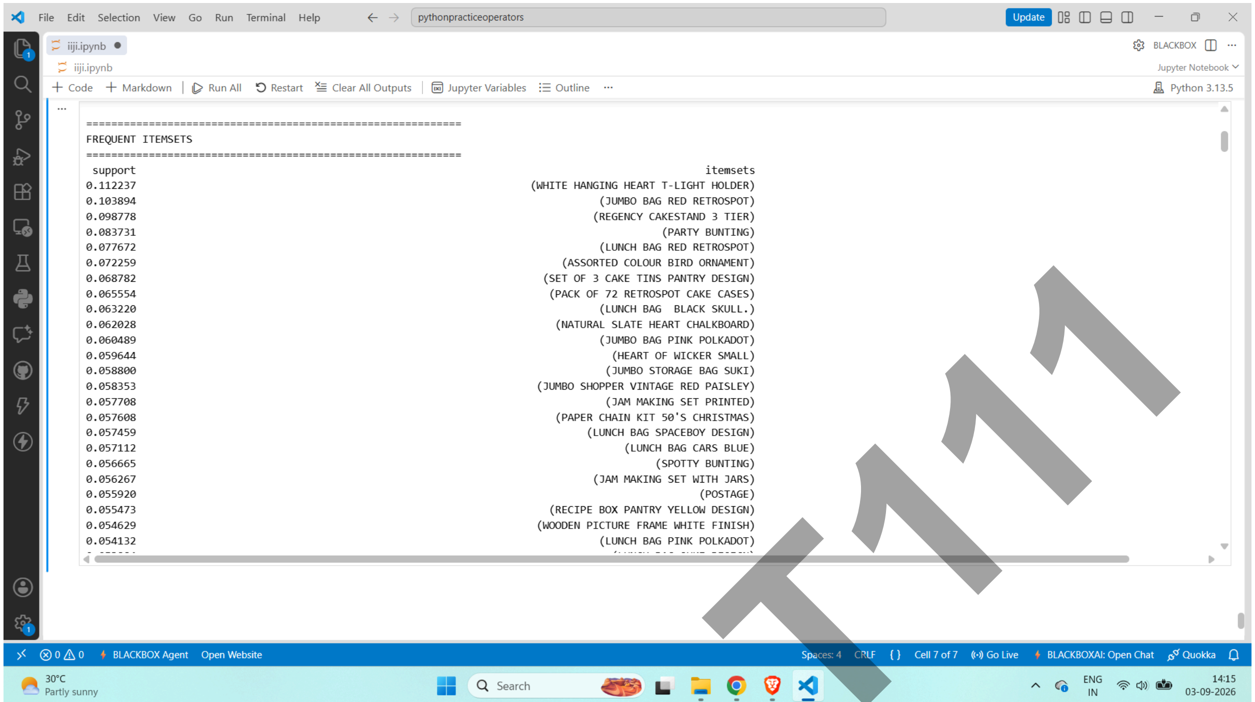
```
print("=" * 60)

if len(rules) > 0:
    print(rules.head(10).to_string(index=False))

# --------------------------------------------------
# Interpret Association Rules
# --------------------------------------------------

print("\n" + "=" * 60)
print("RULE INTERPRETATION")
print("=" * 60)

if len(rules) > 0:
    for _, row in rules.head(10).iterrows():

        antecedent = ", ".join(list(row["antecedents"]))
        consequent = ", ".join(list(row["consequents"]))

        print(
            f"If a customer buys [{antecedent}], "
            f"they are likely to buy [{consequent}]. "
            f"Support = {row['support']:.3f}, "
            f"Confidence = {row['confidence']:.3f}, "
            f"Lift = {row['lift']:.3f}"
        )
else:
    print("No rules available for interpretation.")

# --------------------------------------------------
# Summary
# --------------------------------------------------

print("\n" + "=" * 60)
print("SUMMARY")
print("=" * 60)

print("Number of Transactions:", len(basket))
print("Number of Frequent Itemsets:", len(frequent_itemsets))
```

```python
print("Number of Association Rules:", len(rules))

if len(rules) > 0:
    print("Highest Confidence:",
          round(rules["confidence"].max(), 3))
    print("Highest Lift:",
          round(rules["lift"].max(), 3))

print("\nAssociation Rule Mining using Apriori completed successfully.")
```

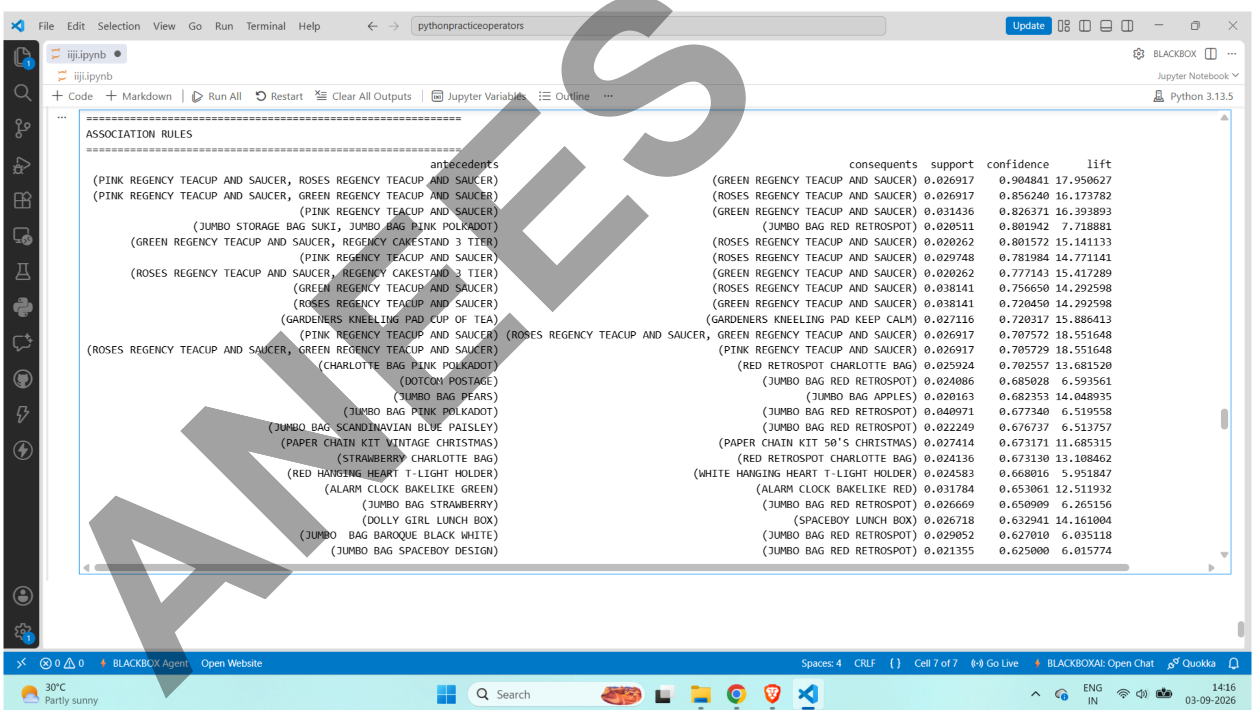
OUTPUT



SHETH L.U.J & SIR M.V COLLEGE OF SCIENCE
SUBJECT : AI



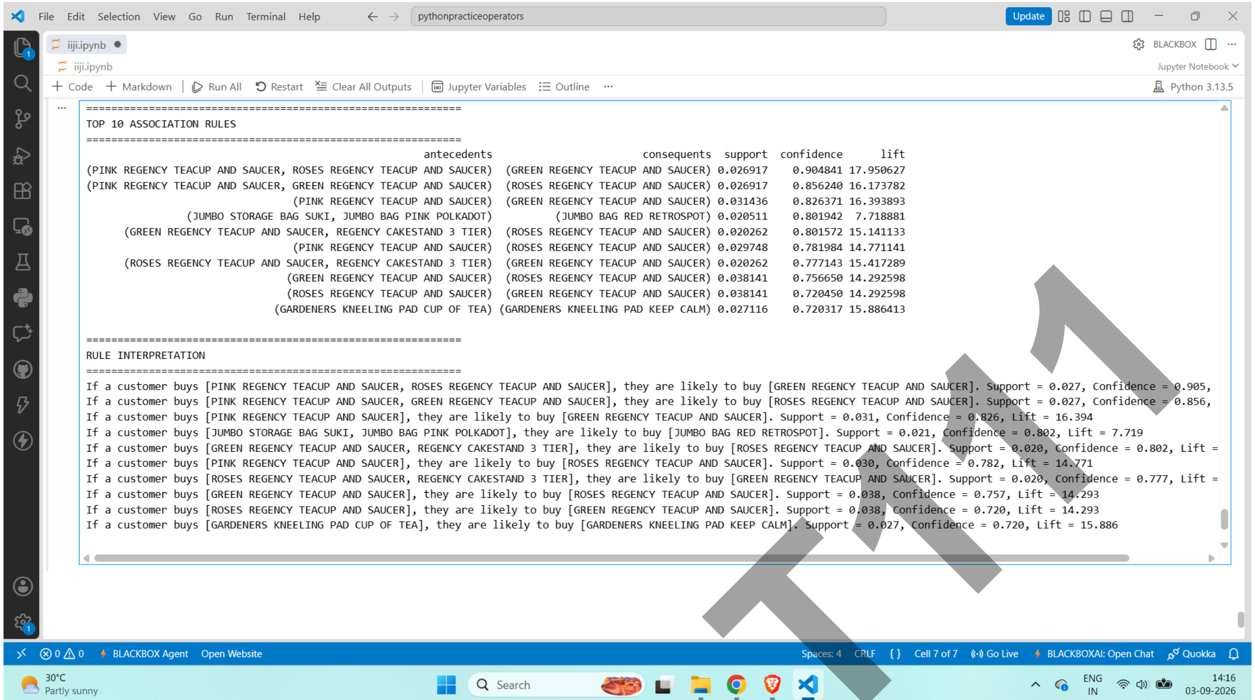
```
============================================================
FREQUENT ITEMSETS
============================================================
 support                                        itemsets
0.112237          (WHITE HANGING HEART T-LIGHT HOLDER)
0.103894                    (JUMBO BAG RED RETROSPOT)
0.098778                     (REGENCY CAKESTAND 3 TIER)
0.083731                              (PARTY BUNTING)
0.077672                    (LUNCH BAG RED RETROSPOT)
0.072259              (ASSORTED COLOUR BIRD ORNAMENT)
0.068782           (SET OF 3 CAKE TINS PANTRY DESIGN)
0.065554            (PACK OF 72 RETROSPOT CAKE CASES)
0.063220                    (LUNCH BAG  BLACK SKULL.)
0.062028             (NATURAL SLATE HEART CHALKBOARD)
0.060489                    (JUMBO BAG PINK POLKADOT)
0.059644                      (HEART OF WICKER SMALL)
0.058800                     (JUMBO STORAGE BAG SUKI)
0.058353          (JUMBO SHOPPER VINTAGE RED PAISLEY)
0.057708                     (JAM MAKING SET PRINTED)
0.057608             (PAPER CHAIN KIT 50'S CHRISTMAS)
0.057459                  (LUNCH BAG SPACEBOY DESIGN)
0.057112                       (LUNCH BAG CARS BLUE)
0.056665                            (SPOTTY BUNTING)
0.056267                    (JAM MAKING SET WITH JARS)
0.055920                                  (POSTAGE)
0.055473             (RECIPE BOX PANTRY YELLOW DESIGN)
0.054629          (WOODEN PICTURE FRAME WHITE FINISH)
0.054132                    (LUNCH BAG PINK POLKADOT)
```



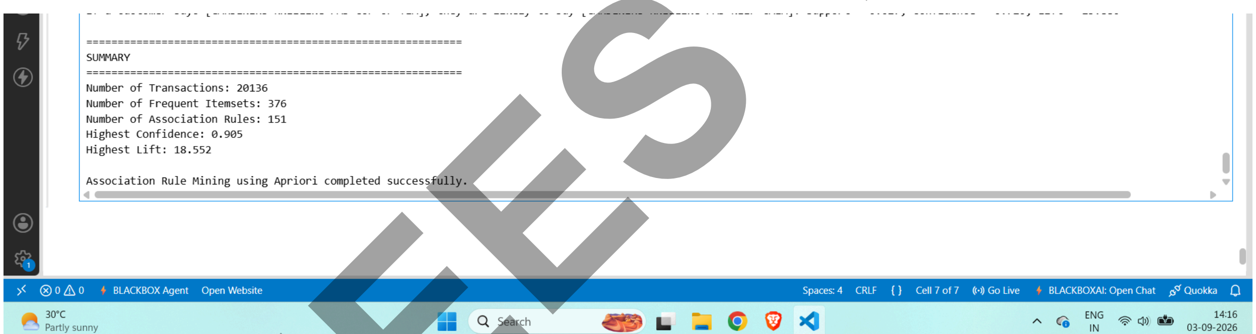
```
============================================================
ASSOCIATION RULES
============================================================
                                          antecedents                                               consequents  support  confidence       lift
 (PINK REGENCY TEACUP AND SAUCER, ROSES REGENCY TEACUP AND SAUCER)                 (GREEN REGENCY TEACUP AND SAUCER) 0.026917    0.904841 17.950627
 (PINK REGENCY TEACUP AND SAUCER, GREEN REGENCY TEACUP AND SAUCER)                 (ROSES REGENCY TEACUP AND SAUCER) 0.026917    0.856240 16.173782
                                  (PINK REGENCY TEACUP AND SAUCER)                 (GREEN REGENCY TEACUP AND SAUCER) 0.031436    0.826371 16.393893
                (JUMBO STORAGE BAG SUKI, JUMBO BAG PINK POLKADOT)                          (JUMBO BAG RED RETROSPOT) 0.020511    0.801942  7.718881
       (GREEN REGENCY TEACUP AND SAUCER, REGENCY CAKESTAND 3 TIER)                 (ROSES REGENCY TEACUP AND SAUCER) 0.020262    0.801572 15.141133
                                  (PINK REGENCY TEACUP AND SAUCER)                 (ROSES REGENCY TEACUP AND SAUCER) 0.029748    0.781984 14.771141
       (ROSES REGENCY TEACUP AND SAUCER, REGENCY CAKESTAND 3 TIER)                 (GREEN REGENCY TEACUP AND SAUCER) 0.020262    0.777143 15.417289
                                 (GREEN REGENCY TEACUP AND SAUCER)                 (ROSES REGENCY TEACUP AND SAUCER) 0.038141    0.756650 14.292598
                                 (ROSES REGENCY TEACUP AND SAUCER)                 (GREEN REGENCY TEACUP AND SAUCER) 0.038141    0.720450 14.292598
                                (GARDENERS KNEELING PAD CUP OF TEA)              (GARDENERS KNEELING PAD KEEP CALM) 0.027116    0.720317 15.886413
                                  (PINK REGENCY TEACUP AND SAUCER) (ROSES REGENCY TEACUP AND SAUCER, GREEN REGENCY TEACUP AND SAUCER) 0.026917    0.707572 18.551648
(ROSES REGENCY TEACUP AND SAUCER, GREEN REGENCY TEACUP AND SAUCER)                  (PINK REGENCY TEACUP AND SAUCER) 0.026917    0.705729 18.551648
                                      (CHARLOTTE BAG PINK POLKADOT)                     (RED RETROSPOT CHARLOTTE BAG) 0.025924    0.702557 13.681520
                                                   (DOTCOM POSTAGE)                         (JUMBO BAG RED RETROSPOT) 0.024086    0.685028  6.593561
                                                 (JUMBO BAG PEARS)                                (JUMBO BAG APPLES) 0.020163    0.682353 14.048935
                                          (JUMBO BAG PINK POLKADOT)                         (JUMBO BAG RED RETROSPOT) 0.040971    0.677340  6.519558
                                (JUMBO BAG SCANDINAVIAN BLUE PAISLEY)                       (JUMBO BAG RED RETROSPOT) 0.022249    0.676737  6.513757
                                 (PAPER CHAIN KIT VINTAGE CHRISTMAS)                 (PAPER CHAIN KIT 50'S CHRISTMAS) 0.027414    0.673171 11.685315
                                           (STRAWBERRY CHARLOTTE BAG)                     (RED RETROSPOT CHARLOTTE BAG) 0.024136    0.673130 13.108462
                                  (RED HANGING HEART T-LIGHT HOLDER)              (WHITE HANGING HEART T-LIGHT HOLDER) 0.024583    0.668016  5.951847
                                        (ALARM CLOCK BAKELIKE GREEN)                       (ALARM CLOCK BAKELIKE RED) 0.031784    0.653061 12.511932
                                              (JUMBO BAG STRAWBERRY)                         (JUMBO BAG RED RETROSPOT) 0.026669    0.650909  6.265156
                                              (DOLLY GIRL LUNCH BOX)                             (SPACEBOY LUNCH BOX) 0.026718    0.632941 14.161004
                                       (JUMBO  BAG BAROQUE BLACK WHITE)                     (JUMBO BAG RED RETROSPOT) 0.029052    0.627010  6.035118
                                            (JUMBO BAG SPACEBOY DESIGN)                     (JUMBO BAG RED RETROSPOT) 0.021355    0.625000  6.015774
```


ANEES T111

SHETH L.U.J & SIR M.V COLLEGE OF SCIENCE
SUBJECT : AI



```
==============================================================
TOP 10 ASSOCIATION RULES
==============================================================
                                             antecedents                          consequents  support  confidence       lift
(PINK REGENCY TEACUP AND SAUCER, ROSES REGENCY TEACUP AND SAUCER)  (GREEN REGENCY TEACUP AND SAUCER) 0.026917    0.904841 17.950627
(PINK REGENCY TEACUP AND SAUCER, GREEN REGENCY TEACUP AND SAUCER)  (ROSES REGENCY TEACUP AND SAUCER) 0.026917    0.856240 16.173782
                                (PINK REGENCY TEACUP AND SAUCER)  (GREEN REGENCY TEACUP AND SAUCER) 0.031436    0.826371 16.393893
                (JUMBO STORAGE BAG SUKI, JUMBO BAG PINK POLKADOT)          (JUMBO BAG RED RETROSPOT) 0.020511    0.801942  7.718881
      (GREEN REGENCY TEACUP AND SAUCER, REGENCY CAKESTAND 3 TIER)  (ROSES REGENCY TEACUP AND SAUCER) 0.020262    0.801572 15.141133
                                (PINK REGENCY TEACUP AND SAUCER)  (ROSES REGENCY TEACUP AND SAUCER) 0.029748    0.781984 14.771141
      (ROSES REGENCY TEACUP AND SAUCER, REGENCY CAKESTAND 3 TIER)  (GREEN REGENCY TEACUP AND SAUCER) 0.020262    0.777143 15.417289
                               (GREEN REGENCY TEACUP AND SAUCER)  (ROSES REGENCY TEACUP AND SAUCER) 0.038141    0.756650 14.292598
                               (ROSES REGENCY TEACUP AND SAUCER)  (GREEN REGENCY TEACUP AND SAUCER) 0.038141    0.720450 14.292598
                             (GARDENERS KNEELING PAD CUP OF TEA) (GARDENERS KNEELING PAD KEEP CALM) 0.027116    0.720317 15.886413

==============================================================
RULE INTERPRETATION
==============================================================
If a customer buys [PINK REGENCY TEACUP AND SAUCER, ROSES REGENCY TEACUP AND SAUCER], they are likely to buy [GREEN REGENCY TEACUP AND SAUCER]. Support = 0.027, Confidence = 0.905,
If a customer buys [PINK REGENCY TEACUP AND SAUCER, GREEN REGENCY TEACUP AND SAUCER], they are likely to buy [ROSES REGENCY TEACUP AND SAUCER]. Support = 0.027, Confidence = 0.856,
If a customer buys [PINK REGENCY TEACUP AND SAUCER], they are likely to buy [GREEN REGENCY TEACUP AND SAUCER]. Support = 0.031, Confidence = 0.826, Lift = 16.394
If a customer buys [JUMBO STORAGE BAG SUKI, JUMBO BAG PINK POLKADOT], they are likely to buy [JUMBO BAG RED RETROSPOT]. Support = 0.021, Confidence = 0.802, Lift = 7.719
If a customer buys [GREEN REGENCY TEACUP AND SAUCER, REGENCY CAKESTAND 3 TIER], they are likely to buy [ROSES REGENCY TEACUP AND SAUCER]. Support = 0.020, Confidence = 0.802, Lift =
If a customer buys [PINK REGENCY TEACUP AND SAUCER], they are likely to buy [ROSES REGENCY TEACUP AND SAUCER]. Support = 0.030, Confidence = 0.782, Lift = 14.771
If a customer buys [ROSES REGENCY TEACUP AND SAUCER, REGENCY CAKESTAND 3 TIER], they are likely to buy [GREEN REGENCY TEACUP AND SAUCER]. Support = 0.020, Confidence = 0.777, Lift =
If a customer buys [GREEN REGENCY TEACUP AND SAUCER], they are likely to buy [ROSES REGENCY TEACUP AND SAUCER]. Support = 0.038, Confidence = 0.757, Lift = 14.293
If a customer buys [ROSES REGENCY TEACUP AND SAUCER], they are likely to buy [GREEN REGENCY TEACUP AND SAUCER]. Support = 0.038, Confidence = 0.720, Lift = 14.293
If a customer buys [GARDENERS KNEELING PAD CUP OF TEA], they are likely to buy [GARDENERS KNEELING PAD KEEP CALM]. Support = 0.027, Confidence = 0.720, Lift = 15.886
```



```
==============================================================
SUMMARY
==============================================================
Number of Transactions: 20136
Number of Frequent Itemsets: 376
Number of Association Rules: 151
Highest Confidence: 0.905
Highest Lift: 18.552

Association Rule Mining using Apriori completed successfully.
```


ANEES T111